

Self-actuated Stretchable Partitions for Dynamically Creating Secure Workplaces

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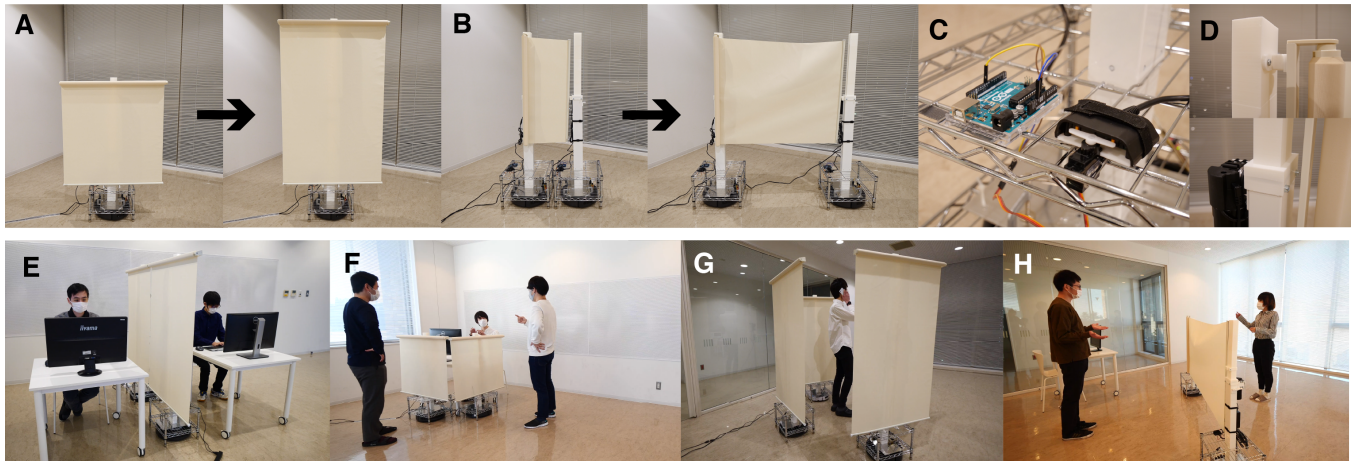


Figure 1: Self-actuated stretchable partition create secure workplaces that protect personal privacy and ensure social distance. Setup: A) Vertically actuating partition for adjusting its height. B) Horizontally actuating partition for adjusting its width. C) Control unit to actuate the height-adjustable stand. D) Roll-up screen mounted on the height-adjustable stand with 3D printed parts. Applications: E) Individual workplace builder, F) Support for informal collaboration, G) Ad-hoc private spot, H) Keeping-distance assistant

ABSTRACT

The configuration of office environments is related to worker satisfaction and can improve work efficiency. Although open offices promote communication among co-workers, privacy issues surface as well as increased risks from such health concerns as viral infections and pandemics. On the other hand, territorial offices protect worker privacy and reduce infection risks. Unfortunately, such arrangements often hinder communication among co-workers. Although a physical office must satisfy different needs under various scenarios with limited space, the lack of flexibility in furniture designs prevents dynamic space management. In this paper, we propose a self-actuated stretchable partition whose physical height,

width, and position can dynamically change to create secure workplaces (e.g., against privacy and infectious risks) without inhibiting group collaboration. To support secure workplace layouts and space reconfigurations, the partitions' height, length, and position are adapted automatically and dynamically. First, we consider the design space of a self-actuated stretchable partition and implement a proof-of concept prototype with a height-adjustable stand, a roll-up screen, and a mobile robot. We then show some example application scenarios to discuss potential future actuated-territorial offices.

CCS CONCEPTS

• **Computer systems organization** → **Human computer interaction (HCI).**

KEYWORDS

Shape-changing Interface; Spatial Configuration; Territorial Office; Privacy; Virus infection; Social distancing

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1 INTRODUCTION

Many people spend a lot of time in a particular office. The spatial configurations of office environments are related to user satisfaction of workplaces [13] and improve individual work performances [6] or productivity. For example, open spaces facilitate communication [2, 5], but they are often noisy [4, 11] and unsafe because such workplaces, without clear territories, are completely exposed to surrounding environment. Workers may easily have privacy and virus propagation concerns, especially under the current circumstances of the COVID-19 pandemic, which might also reduce their productivity. Conversely, territorial offices are more secure and productive [1] because physical partitions effectively protect their desktops and privacy. Such private environments with physical walls ensure sufficient social distancing among workers and can reduce the spread of infectious diseases, which results in protecting the worker's health. Yet, such environments seriously complicate communication with co-workers. Since every office has limited physical space and requires a degree of layout flexibility, most of workplaces select without surrounding by walls. Thus, we are primarily interested in dynamically creating secure and productive workplaces depending on work situations by flexibly arranging multiple partitions.

A conventional approach is to arrange wheeled partitions to create territorial spaces when performing private tasks and remove them for collaborative work. However, manually rearranging static partitions is inefficient and costly. More sophisticated approaches employ virtual partitions using AR glasses to create an ad-hoc individual workplace that effectively blocks the noise from surrounding environments [14]. Weightless Walls is also a virtual partition that creates an invisible individual workplace where the privacy of group conversations is protected by controlling sound transmissions between people using headphones [23]. Although such virtual partitions are quite affordable, their ability to create secure environments is limited (i.e., neither visual privacy or virus protection is ensured), suggesting the need for physical partitions that can flexibly and dynamically change the configuration of an entire space. Thus, researchers have worked on automatic robotic partitions and found that this approach of surrounding personal space motions would provide users with great approachability [15]. However, this robotic partition manipulates its length only, and is unable to change its height and coordinate with multiple partitions to configure different shaped spaces. Thus, while such physical moving partitions would be promising, detailed knowledge to create physically and dynamically secure workplaces is still lacking.

In this work, we propose a novel self-actuated stretchable partition whose physical height, length, and position can be flexibly modified within a space that dynamically creates secure workplaces while allowing group collaboration (see Figure 2, Figure 5). We assume that a certain spatial awareness technology is available to detect the situation around the users. Within this context, each of our partition can change its height, length and position automatically and dynamically to adapt the users' needs. For example, each actuation can be helpful to protect personal privacy, encourage

sufficient social distancing, and ensure necessary transparency for collaboration. Such automatic partitions' arrangement and shape-change would diminish the physical load of manual carrying as well as support users' engagement to their original work. We first consider the design space of a shape-changing partition. To validate our idea, we built a proof-of concept prototype based on our own design space. We implemented two kinds of prototypes: vertical and horizontal stretching partitions. Each was made with a height-adjustable stand, a roll-up screen, and a mobile robot. We also address several promising example application scenarios using our prototypes in an open office context. Lastly, we discuss the current and future potential of territorial offices using automatically actuated partitions. The following are the contributions of this paper:

- We propose a novel self-actuated stretching partition that allows users to dynamically create various types of secure workplaces based on their current requirements.
- We discuss design space for self-actuated stretchable partitions.
- We implement a proof-of concept prototype with affordable materials and design practical use scenarios in an open plan office context and discuss the potential of future actuated-territorial offices.

2 RELATED WORK

2.1 Designing Work Environment

The configuration of work space designs has an effect on people's overall work performance at various level. In the past, work space designs have been classified into two types: open-office and territorial office. In the case of open-offices, each desk is not separated with partitions and every person can observe others' working behavior, which promotes communication [2, 5]. However, surrounding noise from the environment constitute an important drawback. For example, sounds from conversations and people moving around can easily interfere with concentration [4, 11]. Moreover, personal privacy is not protected in this type of office configuration.

In the case of territorial offices, partitions separate each desk so that users may concentrate on work without being distracted by the surrounding environment [1]. This environment is safe in terms of personal privacy, and partitions prevent spreading virus in the air. Therefore, we believe that physical partitions would be basic elements for designing present and future secure and productive office setups. However, as stated, it has critical flexibility issues.

Against this problem, programmable partition systems have been proposed. Weightless Walls protect group conversation by controlling sound transmission between users [23]. The AR partition can also be used to shut out the surrounding environment [14]. Yet, the visual privacy of the user is still exposed in these virtual wall approaches. Therefore, the physical walls is promising approach to solve this problem. Squama is a dimming glass wall-size partition which can partially control its transparency and opacity by applying voltage and simultaneously satisfies the needs for openness and privacy [16], but it does not have physical locomotion capability. Lee et al., confirmed that the behavior of an approaching physical wall induces higher user approachability through an empirical analysis [15]. Unlike our system, this robotic partition does not

support vertical stretching and cannot create different workplace configurations by coordinating multiple robotic partitions.

Moreover, the attention to physical partitions are significantly increased because it is no longer only related to privacy security, but also to the safety of users against risks of virus infection. To go through this pandemic and prepare for future seasonal flu epidemics, our workplaces should have functionality that creates physical barriers or encouraging necessary social distancing to preserve the health of workers.

2.2 Adaptive Furniture

Adaptive interfaces flexibly transform workplaces to suit various task situations. This research field is becoming active in the HCI community.

Firstly, shape-changing displays actively reconfigure the environment through transforming the display depending on tasks and content. Proxemic Transitions [9] is proposed with a design aspect for workplace reconfiguration composed of shape reconfiguration of dynamic furniture as proxemics[8]. Adaptive shape-changing of vertical displays [22] or tabletop displays [10] provide adequate work environments to support both individual and collaborative tasks. In addition, dynamic transforming introduces room-scale interfaces. LivingDesktop interactively actuates the angle and position of a display depending on the direction of working users to support collaboration [3]. Industrial companies also develop automatic moving office chairs [7]. LiftTiles introduces modular inflatable actuators for interactive room-scale prototyping [20]. TilePop uses a similar robotic body-size props, supporting whole-body interactions by using stacked cube-shaped airbags [24]. Thus, the effectiveness of adaptive spatial configuration by automatic furniture or room-scale interface has been proved.

Lastly, manipulation systems for controlling multiple robots support partition control. RoomShift and ZoomWalls consider original algorithms for controlling mobile robots and moving some physical props that provide haptic feedback [19, 25]. ShapeBots proposes a swarm of shape-changing robots which behavior are controlled to represent certain shapes or help users perform tasks [21]. We apply these robot manipulation approaches to controlling partitions.

3 DESIGN SPACE

Before implementation, we explore the design space of automatic self-actuated partition elements.

3.1 Physical Factor

- **Individual form:** A partition is generally formed as a vertical surface. The height is the crucial form factor, which would be decided depending on how much of the user's body would be exposed and how much communication would happen across the partition. In general, it is hard to determine the best height as there is a clear trade-off between privacy protection and ease of communication. For example, high partitions would be preferred for personal offices, yet do not work well for collaborative work. Similarly, the width is also a critical form factor that decides the partition's space segmentation ability. It is also difficult to obtain the best width for each partition as it depends on several dynamic

factors, e.g., number of workers, worker's behavior, tasks, and workplaces. In this work, we focus on controlling these two form factors to dynamically create various workplace configurations.

- **Scalability and workplace transformation pattern:** For indoor scenarios, modular-style partitions with similar forms are quite common for scalability, where multiple partitions can be connected to each other to create various types of space segmentations (e.g., line, L-shape around-desk separations, etc.). Since workplaces can be further altered or become complicated dynamically, specially designed partitions (e.g., curve or dome shape) are not a good solution for open office contexts. Additionally, modular partitions have been used and verified in designing large-scale shape-changing displays [12, 22]. Based on existing practices, we also adopt this design concept of using multiple partitions of the same form to offer reasonable scalability as well as shape-shifting capability.
- **Partition material:** Materials of partitions significantly influence the shielding capability and how well workers can communicate across it. An opaque material is the most common solution to fully shield the objects that need to be concealed. Semi-transparent materials can be used to offer medium awareness while retaining the basic shielding effect. Considering privacy-controllable partitions, roll-up screen, blind screen, and dimming glass [16] materials are all adequate possibilities. Each material has different characteristics and could be used for our purpose. We take the roll-screen approach as it can easily change the critical form factors of height and width of a partition, and it is easily configured using affordable setups (described later).
- **Locomotion capability:** Manually arrangement of wheeled partitions is common approach in offices or public spaces. But, as previously mentioned, it is quite inefficient and sometimes impractical. For achieving dynamic and adaptive workplace reconfiguration, we believe that partitions need to move and rotate dynamically in the space. To give locomotion abilities to partitions, employing robotic actuators for each partition is optimal. Depending on the control mechanism of the actuators (e.g., omnidirectional or two-wheeled), different locomotion capabilities can be achieved, yet safe and simultaneous control of multiple partitions (actuators) in relation to worker movements would be the most important design element. We rely on prior knowledge of swarm robotics [21] to control the position of partitions dynamically.
- **System operation:** There are two operation methods of moving partitions: self manipulation by the user and automatic control. Manipulation by users enables them to decide the partitions' arrangement, and adjust the height and width of each one according to their requirements. In contrast, automatic control would be promising and efficient, but require a sophisticated situational awareness technology. We considered several application scenarios where a Wizard of Oz approach was used to demonstrate the automatic control example.

3.2 Operation Factor

- **Personal privacy:** Individual privacy is linked to work productivity. A high amount of displayed content requires more privacy and physical barriers for hiding the user from the surrounding environment.
- **Collaborative work:** Collaboration is also important in the office environment. Sometimes collaborative work occurs suddenly and transforming the adequate workplace is difficult. In such situations, dynamic partitions support the creation of temporary workplaces.
- **Social distancing:** Keeping enough distance among people is required to prevent infection under pandemics. Partitions help to preserve adequate distances while supporting comfortable collaboration during daily activities. They also constitute barriers to locomotion, so that individual areas can be effectively separated. This promotes the automatic preservation of social distancing.

4 PROTOTYPE IMPLEMENTATION

Based on the design space, we implement the proof-of concept prototype of our self-actuated stretchable partition. To offer multiple types of spatial configurations for supporting individual and group work, the partition's height and length are adjustable. In addition, the partition automatically moves and rotate, and can be placed anywhere.

Figure 1 shows the implemented prototype and system overview. We design two types of prototypes: height or length adjustable partition with vertical or horizontal actuation (Figure 1A,B). The height adjustment of a partition is realized with height-adjustable stands mounted on its metal shelving unit (Figure 1A). The height-adjustable stand pulls up the roll-up screen attached to the top of the stand and the metal unit, up to a maximum height of 200cm, which is enough to hide the face of a standing person. The adjustable length partition consists of a pair of stands mounted on a mobile robot (Roomba, cleaning robot) (Figure 1B). Both sides of a roll-up screen are attached to each stand. For adjusting the length of the wall, a mobile robot moves to extend the screen.

The motor of height-adjustable stand is controlled through serial communication with a PC. The PC also sends commands to the mobile robot using serial communication via Bluetooth for carrying partition. We assume that each partition's position, angles, and height status as well as the nearby users are tracked using a room-scale spatial awareness technology.

5 APPLICATION SCENARIO

Using our prototype, we design practical application scenarios to explore how self-actuated stretchable partitions can arrange a space and dynamically create secure workplaces in open office environments. In the following, we show four promising example applications with two dimensions of partition dynamics (static and dynamic) and context (i.e. privacy and social distancing). Each application does not always strictly satisfy both context, we introduce them to expand for creating productive work environment. An additional spatial awareness system is required to execute each scenario, yet this article focuses on space reconfiguration using our stretching and moving partitions based on a Wizard of Oz approach.

5.1 Individual workplace builder

The first application is an individual workplace builder. Open office environments usually expose the personal workplace of a user. Semi-transparent walls are often used to create explicit territories of each user but unfortunately, communicating with co-workers is difficult in an environment where the individual's desk is surrounded at all times. Figure 2 shows an example situation of an adaptive individual workplace builder. For separating the individual workplaces of users sitting in adjacent seats, the self-actuating partition raise itself automatically. This protects personal privacy and also helps each user to reduce the risk of infection while co-worker is sitting in the next seat.

5.2 Informal Collaboration Supporter

The second application is a switching scenario between individual and collaborative workplaces in an office environment. As Figure 2 shows, hiding a personal desk from its surrounding environment with partitions allows the user to concentrate on their task. The system detects that an informal collaboration happens and the bystanders start approaching to the sitting user, the partition falls to open the user's perspective to communicate with their co-workers. Thus, flexible height adjustment helps to switch between individual and collaborative workplaces instantly and with remaining their original formation.

5.3 Ad-hoc Private Spot

Producing instant private spots with self-actuated partitions means that users can create multi-purpose rooms at anytime and anywhere. Figure 4 shows an example use case of a user answering the phone. When detecting a situation that requires to protect a user's privacy, partitions which were put away in the room gather towards the user and create a personal spot. This dynamic spot can be created instantly when needed, and each partition disperses after finishing the task.

5.4 Keeping-distance Assistant

Lastly, we introduce a guidance application for keeping distance among people Figure 5. Usually, users tend to gather close and keeping enough distance is difficult at the same place. To promote the preservation of greater social distances, partitions automatically approach to the middle of a group of users and adjust the standing area for each of them. Once divided by the partitions, a secure collaborative space for all of users can be created and personal privacy can be protected by hiding their bodies.

6 DISCUSSION AND FUTURE WORK

6.1 Design space

We focused on secure workplace design, specifically addressing personal privacy and infectious risks. They are quite different issues, but interestingly, the solution using physical walls can be shared, which makes us incorporate them into a unified concept of "secure". We will deepen this secure concept in the next design space. We also plan to expand the form and material factors by more referring to air-flow (air-drop) simulations which would help to rethink the future healthy workplace for the age of pandemics.

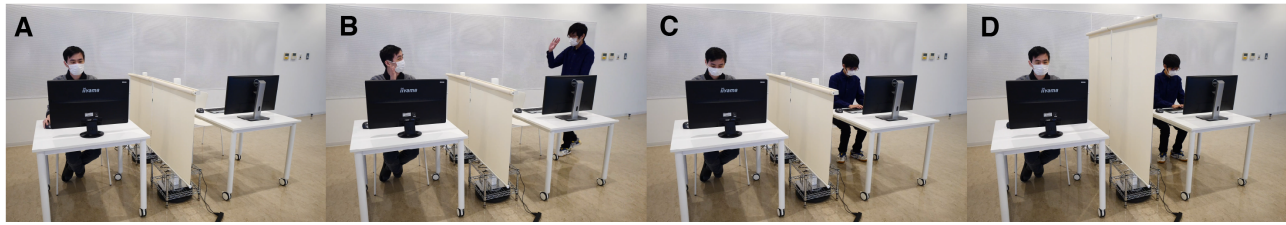


Figure 2: Individual workplace builder: A) A man is working. B) Co-worker returns. C) System detects two users near each other and raises the partition. D) Partition raises to separate their desks and create individual secure workplace.

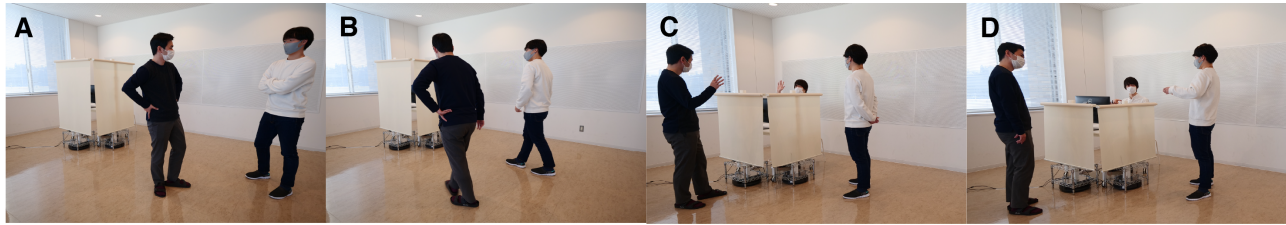


Figure 3: Informal Collaboration Supporter: A) A man is working, and two men are talking near him. B) They approach his work space to discuss together. C) The system detects them and the user is ready to talk, the partitions lowers to allow face-to-face formation. D) They discuss around his desk.

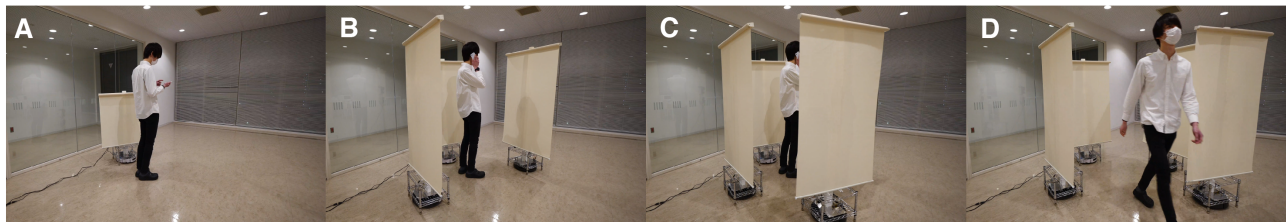


Figure 4: Ad-hoc private spot: A) User enters room, and partition follows him. He stops and answers his phone. B) System detects a private conversation, and other partitions approach. C) They create ad-hoc private spot and surround him during his call. D) When his call is finished, one partition opens like a door, and he walks out. Other partitions also disperse.



Figure 5: Keeping-distance assistant: A) Two users talk at close distance. B) To create greater social distance, partition extends itself between users who notice the approaching partition and stop their conversation. C) They move one step backward, and partition divides their space into two individual areas. D) Users restart their discussion at greater social-distancing positions.

6.2 Hardware Prototyping

Our current hardware is at an early stage with several issues that need to be updated. For example, although the current screen employs affordable roll up screen, it is sufficient for screen stretching but not suitable for screen shrinking. We still explore the best way of dynamically creating secure workplaces. There are also a lot of

future topics for software. Especially with a spatial tracking system and user interface that may allow the users to navigate the moving partitions, a more sophisticated programmable platform is required. As the locomotion actuator, while it has a limited speed and degree of freedom of moving direction, we would keep using the home robot as it is affordable, safe and socially acceptable.

6.3 Situational awareness and User interface

Full automatic partition management is ideal but system also requires to reconcile with the desire of users to retain or manually manage their current environment. Thus, satisfying different requirements of user requires a sophisticated situational awareness system. As such a system has not been achieved, it is difficult to make the system fully automatic at this moment. A reasonable solution is a hybrid control that runs a brief optimization process based on the user's spatial design input. As a user input interface, two options are considered: AR-based egocentric or top-view-based exocentric interface, both can support direct partition manipulations. Based on prior HRI and AR knowledge [17, 18], we will make a novel interface that allows the users to intuitively navigate spatial partitions within the space which is also a new interaction design challenge.

6.4 Use scenarios

We designed four scenarios by combining several design factors in relation to security and social distancing. The future work includes validating if these scenarios are manageable by our partitions and whether our dynamic partitions bring adequate privacy and distancing effects using empirical studies. Since evaluation can be challenging for such dynamic and multi-dimensional applications, We consider dividing the study into several steps to validate the whole concept of the proposed system from different aspects; geometrical factors (e.g., how much the user's body is exposed, etc.), psychological factors (e.g., how safe the user feel), collaborative factors (e.g., how well the users create comfortable formations), and biomedical factors (e.g., how much airdrop lives, by working closely with biomedical researchers).

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REFERENCES

- [1] Hamed S. Alavi, Himanshu Verma, Jakub Mlynar, and Denis Lalanne. 2018. The Hide and Seek of Workspace: Towards Human-Centric Sustainable Architecture. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '18)*. ACM, Article 75, 12 pages. <https://doi.org/10.1145/3173574.3173649>
- [2] Thomas J Allen and Peter G Gerstberger. 1973. A Field Experiment to Improve Communications in a Product Engineering Department: The Nonterritorial Office. *Human Factors* 15, 5 (1973), 487–498.
- [3] Gilles Bailly, Sidharth Sahdev, Sylvain Malacria, and Thomas Pietrzak. 2016. LivingDesktop: Augmenting Desktop Workstation with Actuated Devices. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '16)*. ACM, 5298–5310. <https://doi.org/10.1145/2858036.2858208>
- [4] Simon P Banbury and Dianne C Berry. 2005. Office Noise and Employee Concentration: Identifying Causes of Disruption and Potential Improvements. *Ergonomics* 48, 1 (2005), 25–37.
- [5] Ethan S Bernstein and Stephen Turban. 2018. The Impact of the 'Open' Workspace on Human Collaboration. *Philosophical Transactions of the Royal Society B: Biological Sciences* 373, 1753 (2018), 20170239.
- [6] Aoife Brennan, Jasdeep S Chugh, and Theresa Kline. 2002. Traditional versus Open Office design: A Longitudinal Field Study. *Environment and Behavior* 34, 3 (2002), 279–299.
- [7] Nissan Motor Company. 2016. Intelligent Parking Chair. <http://www.nissan.co.jp/BRAND/TFL/IPC/>.
- [8] Hall T. Edward. 1966. *The Hidden Dimension*. Doubleday.
- [9] Jens Emil Grønbaek, Henrik Korgaard, Marianne Graves Petersen, Morten Henriksen Birk, and Peter Gall Krogh. 2017. Proxemic Transitions: Designing Shape-Changing Furniture for Informal Meetings. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '17)*. ACM, 7029–7041. <https://doi.org/10.1145/3025453.3025487>
- [10] Jens Emil Grønbaek, Majken Kirkegaard Rasmussen, Kim Halskov, and Marianne Graves Petersen. 2020. KirigamiTable: Designing for Proxemic Transitions with a Shape-Changing Tabletop. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM, Article 705, 15 pages. <https://doi.org/10.1145/3313831.3376834>
- [11] Jungsoo Kim and Richard De Dear. 2013. Workspace Satisfaction: The Privacy communication Trade-off in Open-plan Offices. *Journal of Environmental Psychology* 36 (2013), 18–26.
- [12] Yoshiaki Kudo, Kazuki Takashima, Morten Fjeld, and Yoshifumi Kitamura. 2018. AdapTable: Extending Reach over Large Tabletops through Flexible Multi-Display Configuration. In *Proceedings of International Conference on Interactive Surfaces and Spaces (ISS '18)*. ACM, 213–225. <https://doi.org/10.1145/3279778.3279779>
- [13] Virginia Kupritz. 2003. Accommodating Privacy to Facilitate New Ways of Working. *Journal of Architectural and Planning Research* 20, 2 (2003), 122–135.
- [14] Hyelip Lee, Seungwoo Je, Rachel Kim, Himanshu Verma, Hamed Alavi, and Andrea Bianchi. 2019. Partitioning Open-plan Workspaces via Augmented Reality. *Personal and Ubiquitous Computing* (2019), 1–16.
- [15] H. Lee, Y. Kim, and M. Kim. 2013. Come on in!: A Strategic Way to Intend Approachability to a Space by using Motions of a Robotic Partition. In *Proceedings of International Symposium on Robot and Human Interactive Communication (RO-MAN)*. IEEE, 441–446. <https://doi.org/10.1109/ROMAN.2013.6628519>
- [16] Jun Rekimoto. 2012. Squama: Modular Visibility Control of Walls and Windows for Programmable Physical Architectures. In *Proceedings of International Working Conference on Advanced Visual Interfaces (AVI '12)*. ACM, 168–171. <https://doi.org/10.1145/2254556.2254587>
- [17] Daisuke Sakamoto, Koichiro Honda, Masahiko Inami, and Takeo Igarashi. 2009. Sketch and Run: A Stroke-Based Interface for Home Robots. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '09)*. ACM, 197–200. <https://doi.org/10.1145/1518701.1518733>
- [18] Thomas Seifried, Michael Haller, Stacey D. Scott, Florian Perteneder, Christian Rendl, Daisuke Sakamoto, and Masahiko Inami. 2009. CRISTAL: A Collaborative Home Media and Device Controller Based on a Multi-Touch Display. In *Proceedings of International Conference on Interactive Tabletops and Surfaces (ITS '09)*. ACM, 33–40. <https://doi.org/10.1145/1731903.1731911>
- [19] Ryo Suzuki, Hooman Hedayati, Clement Zheng, James L. Bohn, Daniel Szafir, Ellen Yi-Luen Do, Mark D. Gross, and Daniel Leithinger. 2020. RoomShift: Room Scale Dynamic Haptics for VR with Furniture-Moving Swarm Robots. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM, Article 396, 11 pages. <https://doi.org/10.1145/3313831.3376523>
- [20] Ryo Suzuki, Ryosuke Nakayama, Dan Liu, Yasuaki Kakehi, Mark D. Gross, and Daniel Leithinger. 2020. LiftTiles: Constructive Building Blocks for Prototyping Room-Scale Shape-Changing Interfaces. In *Proceedings of International Conference on Tangible, Embedded, and Embodied Interaction (TEI '20)*. ACM, 143–151. <https://doi.org/10.1145/3374920.3374941>
- [21] Ryo Suzuki, Clement Zheng, Yasuaki Kakehi, Tom Yeh, Ellen Yi-Luen Do, Mark D. Gross, and Daniel Leithinger. 2019. ShapeBots: Shape-Changing Swarm Robots. In *Proceedings of Symposium on User Interface Software and Technology (UIST '19)*. ACM, 493–505. <https://doi.org/10.1145/3332165.3347911>
- [22] Kazuki Takashima, Takafumi Oyama, Yusuke Asari, Ehud Sharlin, Saul Greenberg and Yoshifumi Kitamura. 2016. Study and Design of a Shape-Shifting Wall Display. In *Proceedings of Conference on Designing Interactive Systems (DIS '16)*. ACM, 796–806. <https://doi.org/10.1145/2901790.2901892>
- [23] Yuichiro Takeuchi. 2010. Weightless Walls and the Future Office. In *Proceedings of CHI Conference on Human Factors in Computing Systems (CHI '10)*. ACM, 619–628. <https://doi.org/10.1145/1753326.1753419>
- [24] Shan-Yuan Teng, Cheng-Lung Lin, Chi-huan Chiang, Tzu-Sheng Kuo, Liwei Chan, Da-Yuan Huang, and Bing-Yu Chen. 2019. TilePoP: Tile-Type Pop-up Prop for Virtual Reality. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology (UIST '19)*. ACM, 639–649. <https://doi.org/10.1145/3332165.3347958>
- [25] Yan Yixian, Kazuki Takashima, Anthony Tang, Takayuki Tanno, Kazuyuki Fujita and Yoshifumi Kitamura. 2020. ZoomWalls: Dynamic Walls That Simulate Haptic Infrastructure for Room-Scale VR World. In *Proceedings of Symposium on User Interface Software and Technology (UIST '20)*. ACM, 223–235. <https://doi.org/10.1145/3379337.3415859>